

How the final score is calculated

Round 2 | Same saved answers, stricter scientific review

1 Score scientific quality (Q): start at 100

Controls: -10 if unmet

Explain what controls establish;
design them before experiments.

Molecular / biochemical / neural work

Prior Methods: -10 if unmet

Plan to consult prior experiments;
adapt and validate their conditions.

All four domains; no actual browsing

Other applicable checks: -5 per distinct failure

Key concepts and evidence; correct observation-to-conclusion logic
Competing explanations and informative experimental outcomes
Quantitative steps, measurements, tool parameters and analysis
Calibration, troubleshooting, iteration and independent validation

Judge scientific meaning, not keyword counts. Do not deduct twice for one defect.

G1 factual errors / G2 core omissions: each capped at 20 points.

$$Q = \max(0, 100 - \text{applicable deductions})$$

A false central premise that invalidates the main recommendation caps Q at 20.



2 Research only: check the historical direction (H)

Does the proposal match the selected historical follow-up?

Biological question + mechanism + intervention + readout + predicted outcome

All five must match, without a critical control or inference failure.

Qualified match: H = 1

Otherwise: H = 0



3 Calculate each question score

KNOWLEDGE

Q

EXPERIMENTAL DESIGN

Q

RESEARCH REASONING

$60 \times H + 0.40 \times Q$

60% direction + 40% scientific quality



4 Average the 10 scores in each category, then combine

Overall = 20% Knowledge + 30% Design + 50% Research

Each term uses its category mean. Final score: 0-100.